

B3 Atomic and Laser Physics — Model Solutions, Trinity Term 2022

Question 1 — Sr terms, transitions, Zeeman

Problem. (a) LS -coupling, dipole selection rules, terms of $5s^2, 5s5p, 5s6s$ in Sr. (b) Assign 461 nm, 679 nm, 688 nm, 707 nm, 1.12 μm , 689 nm; level diagram in m^{-1} ; identify Normal Zeeman line. (c) $5s5d$ terms; allowed dipole transitions; identify the line splitting into 6 transverse-Zeeman components. (d) Practical apparatus.

(a)

LS -coupling: residual electrostatic $\gg$ spin-orbit, so couple $\mathbf{L} = \sum \mathbf{l}_i$ and $\mathbf{S} = \sum \mathbf{s}_i$ first, then $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$H_{\text{atom}} \approx H_{\text{c.f.}} + H_{\text{res}} + H_{\text{s.o.}}, \quad H_{\text{res}} \gg H_{\text{s.o.}}$$

Good quantum numbers: L, S, J, M_J ; term symbol $^{2S+1}L_J$. Valid for light atoms (small Z , since $H_{\text{s.o.}} \propto Z^4$).

General dipole rules (parity + rank-1 tensor):

$$\Delta J = 0, \pm 1 \ (0 \nleftrightarrow 0), \quad \Delta M_J = 0, \pm 1, \quad \text{parity changes.}$$

LS -specific (dipole acts only on space):

$$\Delta S = 0, \quad \Delta L = 0, \pm 1 \ (0 \nleftrightarrow 0).$$

$5s^2$: equivalent closed sub-shell, Pauli forces $L = S = 0$:

$$^1S_0.$$

$5s5p$: inequivalent, $\ell_1 = 0, \ell_2 = 1, L = 1, S = 0, 1$:

$$^1P_1, \quad ^3P_0, \ ^3P_1, \ ^3P_2.$$

$5s6s$: inequivalent, $\ell_1 = \ell_2 = 0, L = 0, S = 0, 1$:

$$^1S_0, \quad ^3S_1.$$

(b)

Label $g = ^1S_0$ ($5s^2$), $a = ^1P_1$, $b = ^3P_0$, $c = ^3P_1$, $d = ^3P_2$, $e = ^3S_1$, $f = ^1S_0$ ($5s6s$).

461 nm seen in absorption from g : fully allowed singlet-singlet:

$$g \rightarrow a : \ ^1S_0 \rightarrow ^1P_1 \ (461 \text{ nm}).$$

689 nm weak from g : violates $\Delta S = 0$ (intercombination):

$$g \rightarrow c : \ ^1S_0 \rightarrow ^3P_1 \ (689 \text{ nm}).$$

Remaining four lines connect $5s5p \leftrightarrow 5s6s$. Triplet–triplet allowed: ${}^3P_J \rightarrow {}^3S_1$ ($J = 0, 1, 2$). Singlet–singlet allowed: ${}^1P_1 \rightarrow {}^1S_0$. Landé interval rule on 3P orders $J = 0 < 1 < 2$:

$${}^3P_0 \rightarrow {}^3S_1 \text{ (679 nm)}, {}^3P_1 \rightarrow {}^3S_1 \text{ (688 nm)}, {}^3P_2 \rightarrow {}^3S_1 \text{ (707 nm)}, {}^1P_1 \rightarrow {}^1S_0 \text{ (1.12 }\mu\text{m)}.$$

Convert via $\tilde{\nu} = 1/\lambda$ for $g \rightarrow a$:

$$\tilde{\nu}_a = 1/(461 \times 10^{-9} \text{ m}) = 2.169 \times 10^6 \text{ m}^{-1}.$$

For $g \rightarrow c$ (689 nm):

$$\tilde{\nu}_c = 1/(689 \times 10^{-9} \text{ m}) = 1.451 \times 10^6 \text{ m}^{-1}.$$

Convert 688 nm interval:

$$1/(688 \times 10^{-9} \text{ m}) = 1.453 \times 10^6 \text{ m}^{-1}.$$

Add 688 nm interval to $\tilde{\nu}_c$ to reach e :

$$\tilde{\nu}_e = 1.451 \times 10^6 + 1.453 \times 10^6 = 2.904 \times 10^6 \text{ m}^{-1}.$$

Convert 679 nm interval:

$$1/(679 \times 10^{-9} \text{ m}) = 1.473 \times 10^6 \text{ m}^{-1}.$$

Subtract from $\tilde{\nu}_e$ to reach b :

$$\tilde{\nu}_b = 2.904 \times 10^6 - 1.473 \times 10^6 = 1.431 \times 10^6 \text{ m}^{-1}.$$

Convert 707 nm interval:

$$1/(707 \times 10^{-9} \text{ m}) = 1.414 \times 10^6 \text{ m}^{-1}.$$

Subtract from $\tilde{\nu}_e$ to reach d :

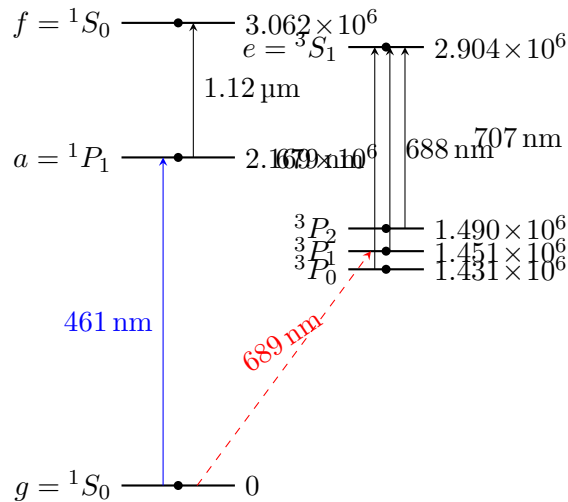
$$\tilde{\nu}_d = 2.904 \times 10^6 - 1.414 \times 10^6 = 1.490 \times 10^6 \text{ m}^{-1}.$$

Convert 1.12 μm interval:

$$1/(1.12 \times 10^{-6} \text{ m}) = 0.893 \times 10^6 \text{ m}^{-1}.$$

Add to $\tilde{\nu}_a$ to reach f :

$$\tilde{\nu}_f = 2.169 \times 10^6 + 0.893 \times 10^6 = 3.062 \times 10^6 \text{ m}^{-1}.$$



Normal Zeeman: needs $g_J = 1$ on both levels, i.e. $S = 0$ on both. Only singlet–singlet line qualifies:

461 nm (${}^1S_0 \rightarrow {}^1P_1$) shows the Normal Zeeman effect.

(c)

$5s5d$: inequivalent, $\ell_1 = 0, \ell_2 = 2, L = 2, S = 0, 1$:

$$^1D_2, \quad ^3D_1, \quad ^3D_2, \quad ^3D_3.$$

Parity of $5s5d$ is even ($\sum \ell_i$ even); same as $5s^2, 5s6s$. Parity rule forbids $5s5d \rightarrow 5s^2, 5s6s$. Allowed only to $5s5p$ (odd). Apply $\Delta S = 0, \Delta L = 0, \pm 1, \Delta J = 0, \pm 1$ ($0 \leftrightarrow 0$):

- $^1D_2 \rightarrow ^1P_1$.
- $^3D_1 \rightarrow ^3P_0, ^3P_1, ^3P_2$.
- $^3D_2 \rightarrow ^3P_1, ^3P_2$.
- $^3D_3 \rightarrow ^3P_2$.

Total: 7 dipole-allowed transitions.

Landé factors (use $g_J = 1 + [J(J+1) + S(S+1) - L(L+1)]/[2J(J+1)]$):

$$g(^3D_1) = 1 + \frac{2+2-6}{4} = \frac{1}{2}.$$

$$g(^3P_1) = 1 + \frac{2+2-2}{4} = \frac{3}{2}.$$

$$g(^3D_2) = 1 + \frac{6+2-6}{12} = \frac{7}{6}.$$

$$g(^3D_3) = 1 + \frac{12+2-6}{24} = \frac{4}{3}.$$

$$g(^3P_2) = 1 + \frac{6+2-2}{12} = \frac{3}{2}.$$

$$g(^1D_2) = g(^1P_1) = 1.$$

Transition shifts in field: $\hbar\omega = \hbar\omega_0 + (g'M' - gM)\mu_B B$, with selection $\Delta M = 0, \pm 1$ (and $0 \rightarrow 0$ forbidden when $\Delta J = 0$). Transverse view shows all ΔM values; count distinct shifts.

Consider $^3D_1 \rightarrow ^3P_1$ ($J' = 1 \rightarrow J = 1, g' = 1/2, g = 3/2$): pairs (M', M) with $|M' - M| \leq 1$ excluding $0 \rightarrow 0$. Enumerate shifts $g'M' - gM$:

$$M' = 1, M = 1 : 1/2 - 3/2 = -1.$$

$$M' = 1, M = 0 : 1/2.$$

$$M' = 0, M = 1 : -3/2.$$

$$M' = 0, M = -1 : 3/2.$$

$$M' = -1, M = 0 : -1/2.$$

$$M' = -1, M = -1 : -1/2 + 3/2 = 1.$$

Collect distinct shifts:

$$\{-3/2, -1, -1/2, +1/2, +1, +3/2\}.$$

Exactly six distinct components:

$^3D_1 \rightarrow ^3P_1 : \text{ splits into 6 components in transverse view.}$

Reasoning: $0 \rightarrow 0$ removal of the central π line at $\Delta = 0$ together with $g' \neq g$ giving paired $\pm$ shifts produces 6 distinct frequencies (no coincidences).

(d)

Need resolution $\gtrsim \mu_B B / h \approx 14 \text{ GHz/T}$, i.e. $\sim 1 \text{ pm}$ at 600 nm . Apparatus:

- Source: Sr hollow-cathode lamp or laser-excited atomic beam.
- Magnet: water-cooled electromagnet, $B \sim 0.1 \text{ T}$ to 1 T , uniform over source.
- Optics: collimation perpendicular to $\mathbf{B}$ (transverse view); polariser to separate π ($\mathbf{E} \parallel \mathbf{B}$) and σ ($\mathbf{E} \perp \mathbf{B}$).
- Spectrometer: Fabry–Pérot etalon (finesse $\gtrsim 30$), FSR matched to splitting; CCD detection.

Issues: Doppler broadening ($\sim 1 \text{ GHz}$ at 600 K) must be smaller than the splitting—use cold atomic beam or laser cooling; calibrate B (Hall probe / NMR); minimise pressure broadening.

Question 2 — Hyperfine interval rule, Eu

Problem. (a) $A \mathbf{I} \cdot \mathbf{J}$ interval rule; physical origins and good quantum numbers. (b) Why $4f^7 6s 6d^8 D_{11/2}$ has large hfs. (c) From peak table, show a, b, c, e obey interval rule, find other ^{151}Eu peaks, deduce $I(^{151}\text{Eu})$. (d) $I(^{153}\text{Eu})$ and μ_I ratio.

(a)

Couple $\mathbf{F} = \mathbf{I} + \mathbf{J}$. Square:

$$\mathbf{F}^2 = \mathbf{I}^2 + \mathbf{J}^2 + 2 \mathbf{I} \cdot \mathbf{J}.$$

Solve for $\mathbf{I} \cdot \mathbf{J}$:

$$\mathbf{I} \cdot \mathbf{J} = \frac{1}{2} (\mathbf{F}^2 - \mathbf{I}^2 - \mathbf{J}^2).$$

Eigenvalue in $|IJFM_F\rangle$:

$$E_F = \frac{1}{2} A \hbar^2 [F(F+1) - I(I+1) - J(J+1)]. \quad (1)$$

Adjacent-level spacing from (1):

$$E_F - E_{F-1} = \frac{1}{2} A \hbar^2 [F(F+1) - (F-1)F].$$

Expand inside the bracket:

$$F(F+1) - (F-1)F = F^2 + F - F^2 + F = 2F.$$

Substitute back:

$$E_F - E_{F-1} = A \hbar^2 F. \quad (2)$$

Spacing $\propto F$: interval rule.

Origins.

- Spin–orbit $\zeta(r) \mathbf{L} \cdot \mathbf{S}$: fine structure within an LS term; good quantum numbers L, S, J, M_J ; energy $\sim 10^{-3} \text{ eV}$ to $\sim 1 \text{ eV}$.
- Magnetic-dipole hyperfine $A \mathbf{I} \cdot \mathbf{J}$: hyperfine multiplet within a level; good quantum numbers I, J, F, M_F ; energy $\sim \mu\text{eV}$ – meV (RF/microwave).

(b)

Two factors:

- $4f^7$ half-filled shell: 7 unpaired spins giving large spin hyperfine field at the nucleus.
- Unpaired 6s electron: $|\psi_{6s}(0)|^2 \neq 0$ produces a large Fermi-contact interaction:

$$A_{6s} \propto |\psi_{6s}(0)|^2 \mu_I.$$

Combined effect makes A unusually large.

(c)

Lower level has negligible hfs, so transition frequencies map to upper-level E_F via (1). Compute pairwise differences along a, b, c, e :

$$\nu_a - \nu_b = 21.19 - 18.37 = 2.82 \text{ GHz.}$$

$$\nu_b - \nu_c = 18.37 - 14.84 = 3.53 \text{ GHz.}$$

$$\nu_c - \nu_e = 14.84 - 10.60 = 4.24 \text{ GHz.}$$

Form ratios with smallest:

$$2.82 : 3.53 : 4.24 = 4 : 5 : 6.$$

Matches (2) with $A' \equiv A\hbar^2/h$ (smallest spacing labelled by lower F of the pair):

$$A' = 2.82/4 = 0.705 \text{ GHz.}$$

Three consecutive spacings of magnitudes $4A', 5A', 6A'$ correspond to four adjacent F levels (the F in $E_F - E_{F-1} = A'F$ takes values 4, 5, 6). The four peaks a, b, c, e therefore correspond to $F \in \{3, 4, 5, 6\}$ in the upper level (ordered with the larger spacings at lower frequency since $A < 0$ here, giving $F_a = 3, F_b = 4, F_c = 5, F_e = 6$).

Predict next interval ($F = 7$ gives spacing $7A'$):

$$\Delta_{67} = 7 \times 0.705 = 4.94 \text{ GHz.}$$

Position of $F = 7$ peak:

$$\nu_e - \Delta_{67} = 10.60 - 4.94 = 5.66 \text{ GHz} \rightarrow i \text{ (5.63).}$$

Predict next ($F = 8$, spacing $8A'$):

$$\Delta_{78} = 8 \times 0.705 = 5.64 \text{ GHz.}$$

Position of $F = 8$ peak:

$$\nu_i - \Delta_{78} = 5.63 - 5.64 \approx 0.0 \rightarrow l.$$

^{151}Eu peaks therefore:

$$\boxed{\{a, b, c, e, i, l\}}.$$

Six adjacent F values ($F = 3$ to $F = 8$) require:

$$2I + 1 = 6.$$

Solve for I :

$$\boxed{I(^{151}\text{Eu}) = \frac{5}{2}}.$$

Check: $F_{\max} = J + I = 11/2 + 5/2 = 8$, $F_{\min} = |J - I| = 11/2 - 5/2 = 3$, consistent.

(d)

Remaining peaks $\{d, f, g, h, j, k\}$ at 10.99, 9.74, 8.17, 6.29, 4.09, 1.58 GHz. Compute consecutive differences:

$$\nu_d - \nu_f = 10.99 - 9.74 = 1.25 \text{ GHz.}$$

$$\nu_f - \nu_g = 9.74 - 8.17 = 1.57 \text{ GHz.}$$

$$\nu_g - \nu_h = 8.17 - 6.29 = 1.88 \text{ GHz.}$$

$$\nu_h - \nu_j = 6.29 - 4.09 = 2.20 \text{ GHz.}$$

$$\nu_j - \nu_k = 4.09 - 1.58 = 2.51 \text{ GHz.}$$

Form ratios:

$$1.25 : 1.57 : 1.88 : 2.20 : 2.51 = 4 : 5 : 6 : 7 : 8.$$

Five spacings labelled 4, 5, 6, 7, 8 imply six adjacent F levels ($F = 3$ to $F = 8$):

$$2I + 1 = 6.$$

Solve for I :

$$I(^{153}\text{Eu}) = \frac{5}{2}.$$

Identify A'_{153} from the smallest spacing ($F = 4$):

$$A'_{153} = 1.25/4 = 0.3125 \text{ GHz.}$$

Same J and same I for both isotopes, so $A \propto \mu_I/I$ with identical I :

$$\frac{A_{151}}{A_{153}} = \frac{\mu_I(^{151}\text{Eu})}{\mu_I(^{153}\text{Eu})}.$$

Substitute the two A' values:

$$\frac{\mu_I(^{151}\text{Eu})}{\mu_I(^{153}\text{Eu})} = \frac{0.705}{0.3125}.$$

Evaluate:

$$\frac{\mu_I(^{151}\text{Eu})}{\mu_I(^{153}\text{Eu})} \approx 2.26.$$

Question 3 — Two-level atom in classical field

Problem. Two-state atom driven by $\hat{D}_0 \cos \omega t$; $\Psi = c_1 \psi_1 e^{-iE_1 t/\hbar} + c_2 \psi_2 e^{-iE_2 t/\hbar}$. (a) Sudden switch-on with $c_1(0) = 1$: find $|c_2|^2 = U \sin^2(\alpha t)$ under $|\omega - \omega_0| \ll \omega_0$, give U, α , and $\tau \rightarrow \infty$ time average. (b) Rabi vs. Einstein. (c) Lorentzian from natural broadening (Fourier); another mechanism. (d) Pulse $\hat{D}_0 e^{-b|t|} \cos \omega t$, weak field: $c_2(t > 0)$, $|c_2(\infty)|^2$, FWHM.

(a)

Project TDSE onto $\psi_{1,2}$ using $\langle \psi_i | \hat{D}_0 | \psi_i \rangle = 0$:

$$i\hbar \dot{c}_2 = D_{21} \cos(\omega t) e^{i\omega_0 t} c_1.$$

$$i\hbar \dot{c}_1 = D_{12} \cos(\omega t) e^{-i\omega_0 t} c_2.$$

Expand the cosine:

$$\cos \omega t = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t}).$$

Apply RWA (drop $e^{\pm i(\omega+\omega_0)t}$, valid for $|\omega - \omega_0| \ll \omega_0$):

$$i\hbar \dot{c}_2 = \frac{1}{2} D_{21} e^{-i\delta t} c_1, \quad i\hbar \dot{c}_1 = \frac{1}{2} D_{12} e^{i\delta t} c_2, \quad \delta = \omega - \omega_0. \quad (3)$$

Define on-resonance Rabi frequency:

$$\Omega_0 \equiv |D_{21}|/\hbar.$$

Differentiate the c_2 equation in time:

$$i\hbar \ddot{c}_2 = \frac{1}{2} D_{21} (-i\delta) e^{-i\delta t} c_1 + \frac{1}{2} D_{21} e^{-i\delta t} \dot{c}_1.$$

Substitute $\dot{c}_1$ from (3):

$$i\hbar \ddot{c}_2 = -i\delta i\hbar \dot{c}_2 + \frac{1}{2} D_{21} e^{-i\delta t} \cdot \frac{D_{12} e^{i\delta t} c_2}{2i\hbar}.$$

Simplify the second term using $|D_{21}|^2 = \hbar^2 \Omega_0^2$:

$$i\hbar \ddot{c}_2 = -i\delta i\hbar \dot{c}_2 - \frac{i\hbar \Omega_0^2}{4} c_2.$$

Divide through by $i\hbar$:

$$\ddot{c}_2 + i\delta \dot{c}_2 + \frac{1}{4} \Omega_0^2 c_2 = 0.$$

Initial conditions: $c_2(0) = 0$ and $\dot{c}_2(0) = -i\Omega_0/2$ from (3) with $c_1(0) = 1$. Try $c_2(t) = e^{-i\delta t/2} f(t)$:

$$\ddot{f} + \frac{1}{4}(\Omega_0^2 + \delta^2) f = 0.$$

Define generalised Rabi frequency:

$$\Omega \equiv \sqrt{\Omega_0^2 + \delta^2}.$$

General solution $f = A \cos(\Omega t/2) + B \sin(\Omega t/2)$; impose $f(0) = 0$:

$$A = 0.$$

Impose $\dot{f}(0) = \dot{c}_2(0) = -i\Omega_0/2$ (since $f(0) = 0$):

$$B\Omega/2 = -i\Omega_0/2 \implies B = -i\Omega_0/\Omega.$$

Combine:

$$c_2(t) = -\frac{i\Omega_0}{\Omega} e^{-i\delta t/2} \sin(\frac{1}{2}\Omega t). \quad (4)$$

Take modulus squared:

$$|c_2(t)|^2 = \frac{\Omega_0^2}{\Omega_0^2 + \delta^2} \sin^2(\frac{1}{2}\Omega t).$$

Read off U and α :

$$U = \frac{\Omega_0^2}{\Omega_0^2 + \delta^2}, \quad \alpha = \frac{1}{2} \sqrt{\Omega_0^2 + (\omega - \omega_0)^2}.$$

Use $\sin^2(\alpha t) = \frac{1}{2}(1 - \cos 2\alpha t)$:

$$\overline{|c_2|^2} = \frac{U}{\tau} \int_0^\tau \sin^2(\alpha t) dt.$$

Evaluate the integral:

$$\overline{|c_2|^2} = \frac{U}{2} \left[1 - \frac{\sin(2\alpha\tau)}{2\alpha\tau} \right].$$

Take $\tau \rightarrow \infty$ (oscillatory term vanishes):

$$\lim_{\tau \rightarrow \infty} \overline{|c_2|^2} = \frac{U}{2} = \frac{\Omega_0^2}{2(\Omega_0^2 + \delta^2)}.$$

(b)

Rabi oscillations require:

- Coherent monochromatic field (ω well defined).
- Rabi period $\ll$ decoherence time: $\Omega T_2 \gg 1$.
- Interaction time $\gtrsim 1/\Omega$.

Einstein rate equations apply when:

- Broad-spectrum incoherent radiation, $u(\omega)$ slowly varying over linewidth.
- Decoherence dominates: $\Omega \ll 1/T_2$, i.e. phase memory destroyed between events.
- Recovered by adiabatic elimination of coherences in the density-matrix description.

(c)

Decaying upper-state amplitude:

$$\psi(t) = \psi_0 e^{-i\omega_0 t} e^{-\Gamma t/2} \Theta(t), \quad \Gamma = 1/\tau.$$

Fourier transform:

$$\tilde{\psi}(\omega) = \int_0^\infty \psi_0 e^{i\omega t} e^{-i\omega_0 t} e^{-\Gamma t/2} dt.$$

Combine exponents:

$$\tilde{\psi}(\omega) = \psi_0 \int_0^\infty e^{[i(\omega - \omega_0) - \Gamma/2]t} dt.$$

Evaluate the integral:

$$\tilde{\psi}(\omega) = \frac{\psi_0}{\Gamma/2 - i(\omega - \omega_0)}.$$

Take modulus squared:

$$|\tilde{\psi}(\omega)|^2 = \frac{|\psi_0|^2}{(\Gamma/2)^2 + (\omega - \omega_0)^2}.$$

Read off Lorentzian profile:

$$I(\omega) \propto \frac{1}{(\omega - \omega_0)^2 + (\Gamma/2)^2} \text{ (Lorentzian, FWHM} = \Gamma).$$

Another Lorentzian mechanism: **collisional (pressure) broadening**. Random phase-interrupting collisions truncate the radiating dipole at exponential intervals; the autocorrelation decays as $e^{-\Gamma_{\text{coll}} t}$, giving a Lorentzian of width $\Gamma_{\text{coll}} = 2/\tau_{\text{coll}}$.

(d)

First-order perturbation with $c_1 \approx 1$ and $c_2(-\infty) = 0$:

$$c_2(t) = \frac{1}{i\hbar} \int_{-\infty}^t D_{21} e^{-b|t'|} \cos(\omega t') e^{i\omega_0 t'} dt'.$$

Apply RWA (keep $e^{-i\omega t'}$ piece of the cosine):

$$c_2(t) = \frac{D_{21}}{2i\hbar} \int_{-\infty}^t e^{-b|t'|} e^{i(\omega_0 - \omega)t'} dt'.$$

Define detuning $\delta = \omega - \omega_0$:

$$c_2(t) = \frac{D_{21}}{2i\hbar} \int_{-\infty}^t e^{-b|t'|} e^{-i\delta t'} dt'.$$

Split the integral at $t' = 0$ for $t > 0$:

$$c_2(t) = \frac{D_{21}}{2i\hbar} \left[\int_{-\infty}^0 e^{(b-i\delta)t'} dt' + \int_0^t e^{-(b+i\delta)t'} dt' \right].$$

Evaluate each integral:

$$c_2(t) = \frac{D_{21}}{2i\hbar} \left[\frac{1}{b-i\delta} + \frac{1 - e^{-(b+i\delta)t}}{b+i\delta} \right]. \quad (5)$$

Take $t \rightarrow \infty$ (second exponential vanishes since $b > 0$):

$$c_2(\infty) = \frac{D_{21}}{2i\hbar} \left[\frac{1}{b-i\delta} + \frac{1}{b+i\delta} \right].$$

Add the fractions:

$$c_2(\infty) = \frac{D_{21}}{2i\hbar} \cdot \frac{(b+i\delta) + (b-i\delta)}{b^2 + \delta^2}.$$

Simplify the numerator:

$$c_2(\infty) = \frac{D_{21}}{2i\hbar} \cdot \frac{2b}{b^2 + \delta^2} = \frac{D_{21}}{i\hbar} \cdot \frac{b}{b^2 + \delta^2}.$$

Take modulus squared:

$$|c_2(\infty)|^2 = \frac{|D_{21}|^2}{\hbar^2} \cdot \frac{b^2}{(b^2 + \delta^2)^2}.$$

This is a squared Lorentzian in δ .

Maximum at $\delta = 0$:

$$|c_2(\infty)|_{\text{max}}^2 = \frac{|D_{21}|^2}{\hbar^2 b^2}.$$

Half-max condition $|c_2|^2 = \frac{1}{2}|c_2|_{\max}^2$:

$$\frac{b^2}{(b^2 + \delta_{1/2}^2)^2} = \frac{1}{2b^2}.$$

Rearrange:

$$(b^2 + \delta_{1/2}^2)^2 = 2b^4.$$

Take positive square root:

$$b^2 + \delta_{1/2}^2 = b^2\sqrt{2}.$$

Solve for $\delta_{1/2}^2$:

$$\delta_{1/2}^2 = b^2(\sqrt{2} - 1).$$

FWHM is $2|\delta_{1/2}|$:

$$\text{FWHM} = 2b\sqrt{\sqrt{2} - 1} \approx 1.287b.$$

Question 4 — Saturated gain and Na D2 absorption

Problem. Four-level homogeneous transition, pump R_2 , decay A_{21} , lower-level lifetime τ_1 . (a) Rate equations; show $\alpha = \alpha_0/(1 + I/I_s)$, give α_0, I_s . (b) Amplifier integral. (c) Two amplifiers $I_0 \rightarrow 70I_0 \rightarrow 150I_0$, find I_0/I_s . (d) Two-level absorber; Na D2 cross-section from given transmissions; homogeneous broadening conditions.

(a)

Photon flux $I/(\hbar\omega)$; population inversion $\Delta N \equiv N_2 - (g_2/g_1)N_1$. Rate equations:

$$\dot{N}_2 = R_2 - A_{21}N_2 - \sigma \frac{I}{\hbar\omega} \Delta N, \tag{6}$$

$$\dot{N}_1 = A_{21}N_2 - \frac{N_1}{\tau_1} + \sigma \frac{I}{\hbar\omega} \Delta N. \tag{7}$$

Add (6)+(7) in steady state:

$$0 = R_2 - N_1/\tau_1.$$

Solve for N_1 :

$$N_1 = R_2\tau_1.$$

Set $\dot{N}_2 = 0$ in (6) and substitute $N_1 = R_2\tau_1$ into ΔN :

$$0 = R_2 - A_{21}N_2 - \sigma \frac{I}{\hbar\omega} (N_2 - (g_2/g_1)R_2\tau_1).$$

Group N_2 terms:

$$(A_{21} + \sigma I/(\hbar\omega))N_2 = R_2 + (g_2/g_1)R_2\tau_1\sigma I/(\hbar\omega).$$

Solve for N_2 :

$$N_2 = \frac{R_2 + (g_2/g_1)R_2\tau_1\sigma I/(\hbar\omega)}{A_{21} + \sigma I/(\hbar\omega)}.$$

Form $\Delta N = N_2 - (g_2/g_1)R_2\tau_1$:

$$\Delta N = \frac{R_2 + (g_2/g_1)R_2\tau_1\sigma I/(\hbar\omega) - (g_2/g_1)R_2\tau_1(A_{21} + \sigma I/(\hbar\omega))}{A_{21} + \sigma I/(\hbar\omega)}.$$

Cancel the $\sigma I/(\hbar\omega)$ pieces in the numerator:

$$\Delta N = \frac{R_2 - (g_2/g_1)R_2\tau_1 A_{21}}{A_{21} + \sigma I/(\hbar\omega)}.$$

Factor A_{21} in the denominator:

$$\Delta N = \frac{(R_2/A_{21})[1 - (g_2/g_1)A_{21}\tau_1]}{1 + \sigma I/(\hbar\omega A_{21})}.$$

Identify $\alpha = \sigma\Delta N$:

$$\boxed{\alpha = \frac{\alpha_0}{1 + I/I_s}, \quad \alpha_0 = \frac{\sigma R_2}{A_{21}} \left[1 - \frac{g_2}{g_1} A_{21} \tau_1 \right], \quad I_s = \frac{\hbar\omega A_{21}}{\sigma}.}$$

α_0 : small-signal gain. I_s : stimulated rate equals spontaneous rate; gain halved at $I = I_s$.

(b)

Beer-Lambert with saturated gain:

$$\frac{dI}{dz} = \alpha I = \frac{\alpha_0 I}{1 + I/I_s}.$$

Rearrange to separate variables:

$$\frac{1 + I/I_s}{I} dI = \alpha_0 dz.$$

Split the fraction:

$$\left(\frac{1}{I} + \frac{1}{I_s} \right) dI = \alpha_0 dz.$$

Integrate $z \in [0, L]$, $I \in [I_{\text{in}}, I_{\text{out}}]$:

$$\boxed{\ln\left(\frac{I_{\text{out}}}{I_{\text{in}}}\right) + \frac{I_{\text{out}} - I_{\text{in}}}{I_s} = \alpha_0 L.}$$

(c)

Let $x = I_0/I_s$. Stage 1 ($I_0 \rightarrow 70I_0$):

$$\ln 70 + (70 - 1)x = \alpha_0 L.$$

Simplify the coefficient:

$$\ln 70 + 69x = \alpha_0 L.$$

Stage 2 ($70I_0 \rightarrow 150I_0$):

$$\ln(150/70) + (150 - 70)x = \alpha_0 L.$$

Simplify the coefficient:

$$\ln(15/7) + 80x = \alpha_0 L.$$

Set the two RHS equal (both equal $\alpha_0 L$):

$$\ln 70 + 69x = \ln(15/7) + 80x.$$

Collect x on the right and logs on the left:

$$\ln 70 - \ln(15/7) = 11x.$$

Combine logs using $\ln 70 - \ln(15/7) = \ln(70 \cdot 7/15)$:

$$\ln(490/15) = 11x.$$

Evaluate the log:

$$\ln(32.667) = 3.487.$$

Solve for x :

$$\boxed{\frac{I_0}{I_s} = \frac{3.487}{11} \approx 0.317.}$$

(d)

Two-level absorber: drop pump R_2 , keep $N_1 + N_2 = N_0$. Steady-state rate equation:

$$\dot{N}_2 = -A_{21}N_2 + \sigma_{\text{abs}} \frac{I}{\hbar\omega} \left(N_1 - \frac{g_2}{g_1} N_2 \right) = 0.$$

Use $N_1 = N_0 - N_2$ and solve for N_2 :

$$N_2 = \frac{\sigma_{\text{abs}} I / (\hbar\omega) N_0}{A_{21} + \sigma_{\text{abs}} I / (\hbar\omega) (1 + g_2/g_1)}.$$

Form the absorption coefficient $\alpha_{\text{abs}} = \sigma_{\text{abs}}(N_1 - (g_2/g_1)N_2)$ and simplify:

$$\alpha_{\text{abs}} = \frac{\sigma_{\text{abs}} N_0}{1 + I/I_{\text{sat}}}.$$

Saturation intensity:

$$I_{\text{sat}} = \frac{\hbar\omega A_{21}}{\sigma_{\text{abs}}(1 + g_2/g_1)}.$$

Beer-Lambert with $-dI/dz = \alpha_{\text{abs}} I$ integrates to:

$$-\ln(I_{\text{out}}/I_{\text{in}}) - \frac{I_{\text{out}} - I_{\text{in}}}{I_{\text{sat}}} = \sigma_{\text{abs}} N_0 L. \quad (8)$$

Low-intensity limit drops the second term:

$$-\ln 0.90 = \sigma_{\text{abs}} N_0 L.$$

Strong-beam case $I_{\text{out}}/I_{\text{in}} = 0.967$:

$$-\ln 0.967 - \frac{(0.967 - 1)I}{I_{\text{sat}}} = \sigma_{\text{abs}} N_0 L.$$

Substitute $\sigma_{\text{abs}} N_0 L = -\ln 0.90$:

$$-\ln 0.967 + \frac{0.033 I}{I_{\text{sat}}} = -\ln 0.90.$$

Rearrange for the saturation term:

$$\frac{0.033 I}{I_{\text{sat}}} = \ln 0.967 - \ln 0.90.$$

Combine logs:

$$\frac{0.033 I}{I_{\text{sat}}} = \ln(0.967/0.90).$$

Evaluate the log:

$$\ln(1.0744) = 0.0718.$$

Solve for I_{sat} :

$$I_{\text{sat}} = \frac{0.033 \times 270}{0.0718}.$$

Evaluate:

$$I_{\text{sat}} = \frac{8.91}{0.0718} \approx 124 \text{ Wm}^{-2}.$$

Na D2: $g_1 = 2$ ($^2S_{1/2}$), $g_2 = 4$ ($^2P_{3/2}$):

$$1 + g_2/g_1 = 3.$$

Cross-section from I_{sat} using $A_{21} = 1/\tau_{3p}$:

$$\sigma_{\text{abs}} = \frac{\hbar\omega A_{21}}{(1 + g_2/g_1)I_{\text{sat}}} = \frac{\hbar\omega}{3\tau_{3p}I_{\text{sat}}}.$$

Photon energy:

$$\hbar\omega = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3 \times 10^8)}{589 \times 10^{-9}}.$$

Evaluate:

$$\hbar\omega = 3.37 \times 10^{-19} \text{ J}.$$

Plug $\tau_{3p} = 16 \text{ ns}$, $I_{\text{sat}} = 124 \text{ Wm}^{-2}$:

$$\sigma_{\text{abs}} = \frac{3.37 \times 10^{-19}}{3 \times 16 \times 10^{-9} \times 124}.$$

Evaluate the denominator:

$$3 \times 16 \times 10^{-9} \times 124 = 5.95 \times 10^{-6}.$$

Divide:

$$\sigma_{\text{abs}} \approx \frac{3.37 \times 10^{-19}}{5.95 \times 10^{-6}} \approx 5.7 \times 10^{-14} \text{ m}^2.$$

Compare with the resonant limit $\lambda^2/(2\pi)$:

$$(589 \times 10^{-9})^2/(2\pi) \approx 5.5 \times 10^{-14} \text{ m}^2 \checkmark.$$

Resonant cross-section is $\sim \lambda^2/(2\pi)$, $\sim 10^6 \times$ geometrical $\pi a_0^2 \sim 10^{-20} \text{ m}^2$, and $\sim 10^4\text{--}10^5 \times$ collisional $\sigma_{\text{coll}} \sim 10^{-19} \text{ m}^2$.

Homogeneous broadening: every atom has the same line shape, $\Gamma_{\text{hom}} \gtrsim \Gamma_{\text{inhom}}$. For Na, $\Gamma_{\text{nat}}/(2\pi) = 1/(2\pi\tau_{3p}) \approx 10 \text{ MHz}$, but Doppler $\Gamma_D/(2\pi) \sim 1 \text{ GHz}$ at room temperature. Conditions:

- Laser cooling: $\Gamma_D \ll \Gamma_{\text{nat}}$ (cold-atom sample here).
- Sub-Doppler spectroscopy (saturation, two-photon Doppler-free).
- Collisional/pressure regime where $\Gamma_{\text{coll}} \gg \Gamma_D$.